Rates of descent were determined by obtaining the difference, to the nearest 100 feet, between the altitude where the steepest portion of the run began and the altitude at pullout and dividing this difference by the lapsed time in seconds. Since flaps were sometimes used to slow the descent rate during the retardant runs, values of indicated airspeed were also read at the start of the steepest rate of descent and at pullout, for comparison with the design flap speed.

Flight duration, time from lift-off to first drop, and time from last drop to touchdown, were read to the nearest minute. The number of retardant drops per flight was determined and the time between drops for given flights was measured to the nearest quarter of a minute.

Acceleration data were read in 0.1g intervals. Only the maximum positive acceleration recorded during pullout from a retardant drop and the maximum negative acceleration that occured, either at pushover to begin the run, or in rounding out after the pullout, were read. These acceleration data were cumulated by number of runs from each aircraft for comparison with each other and with the design maneuver limit and ultimate load factors. The acceleration data from both aircraft were also combined and cumulated by the frequency of occurrence per hour of flight for comparison with maneuver accelerations experienced by the same type of airplane flown in commercial transport operations.

RESULTS AND DISCUSSION

The variation in flight duration of 415 flights made by two retardant bombers used in forest firefighting operations is shown in figure 1. The shortest flight was 12 minutes in duration, and the longest was 147 minutes. The average flight lasted 49 minutes and 68.3 percent of the flights, or the one-sigma value, were between 28 and 70 minutes in duration.

Elapsed time from lift-off to the first retardant drop for 191 flights is shown in figure 2. Drops were made from as soon as 4 minutes to as long as 65 minutes after lift-off. The majority of the drops occurred from 10 to 40 minutes after lift-off and the average elapsed time was 25 minutes.

The variation in dash speed to the first drop is given in figure 3. Dash speeds measured during 191 flights ranged from 143 to 195 knots. The average dash speed was 168 knots. The relationship of time and dash speed to the first retardant drop is shown in table II. In flights to fires that require less than 20 minutes to reach, dash speeds of up to 180 knots were used. For flights in which more than 20 minutes were needed to reach the fire, dash speeds of up to 195 knots were used. In general, table II indicates that for a given time interval required to reach a fire, a rather broad range of dash speeds is used.